09 — Negative Results, Limitations, and Open Problems

SEION-KGR v26 MAX

Campaign ID: gate12-closeout-2026-08-01
Canonical commit: `6af3c35271ae2ffab41ecba2aad098d1988fdc0c`
Branch: `campaign/gate12-closeout`
Date: 2026-08-01

Purpose: state plainly what did not work, what remains unresolved, and what must not be claimed. This document exists so that the strongest-looking result elsewhere in this package cannot be read in isolation.

Contents

1 Hypotheses not supported by this campaign's evidence

No hypothesis from `preregistration.md` (H1–H4) is confirmed or refuted by confirmatory evidence in this campaign — none was collected (§6). What follows are honestly labeled screening/exploratory/engineering findings only.

2 The path-reasoner residual gate did not open in a real training run

A 40-epoch run on a small synthetic structured graph (16 entities, TuckER base + path reasoner, dim 12) left the path branch's residual gate $\sigma(\gamma_r)$ at essentially its near-zero init value (0.0180 vs. init 0.0180) — the path branch had negligible measured influence on the final score throughout training. Consequently, a queried-edge-leakage negative control on the same setup showed a null effect size (leaky MRR = clean MRR exactly, 0.2961), which technically satisfies "did not get worse" but is a weak, non-differentiating result, investigated rather than accepted at face value (`campaigns/gate12/negative_controls_results.json`).

This is flagged as assumption A25 (OPEN) in the assumption ledger. It is a genuine open question for any future Gate 12 confirmatory campaign: does the near-zero-init residual router open meaningfully within realistic training budgets on real benchmark data, or does the gate need a separate learning-rate schedule, a warm-start, or a different initialization to ever contribute? If the gate systematically fails to open, the entire premise of "start as baseline, only grow if helpful" collapses into "never grows" — indistinguishable from the branch not existing at all. This must be checked before any claim that the path/seionic/E8 branches improve MRR.

3 The path reasoner does not scale to full-dataset batches within this session's compute budget

A live timing probe (A3 configuration: TuckER + path reasoner + seionic branch, full WN18RR, batch_size 256, 1 epoch, GPU) ran over 8 minutes without completing training a single epoch, and was stopped. A CPU retry with a 5,000-triple subsample and batch_size 128 was used instead for Tier 0's A3 ladder point (§6); see `preregistration.md` §11 for the logged deviation. The root cause is architectural, not a bug: `PathReasoner.run_batch_frontiers` is a per-sample Python loop (BFS frontier expansion), which does not vectorize across the batch dimension the way the base experts' pure-GEMM scoring does. **This is a real, unresolved engineering limitation, not a smoke-test inconvenience** — any future Gate 12 confirmatory campaign on full FB15K-237/WN18RR with the path reasoner enabled needs either a batched/vectorized BFS reimplementation or a much longer compute budget (hours, not minutes, per configuration) before it can be run at the scale the mandate specifies.

3.1 GPU was not faster for this workload

The same A3 configuration on GPU (kernel-launch-overhead-bound, many tiny per-sample tensor ops) did not complete within the same wall-clock budget where a CPU run made visible progress. This is consistent with, not contrary to, the finite-tree track's own prior finding (measured 3.2–3.5× GPU slowdown for small operations in a 208-configuration sweep, `numerical_study`) and the mandate's own explicit warning against assuming GPU always wins. Device selection for path-reasoner-enabled runs should default to CPU, or a properly batched GPU-friendly reimplementation is needed — neither is done in this campaign.

4 Certification coverage is honestly near-zero for any compressed model

§**??**(referencing 07): the certification chain (Phase B4) only ever certifies the case where no rank-reducing projector and no nonlinear envelope are in the score path. For any model with the path reasoner enabled (which always introduces the LayerNorm+tanh envelope), certified coverage is exactly 0% — not a bug, the honest and correct output of a chain that refuses to certify what it cannot check. Closing this gap requires either a checked operator-norm bound on closure leakage (assumption A4, still not granted anywhere in this repository) or a checked Lipschitz constant for the nonlinear envelope (A6, likewise not granted). Neither is solved in this campaign.

5 E8's causal contribution remains entirely open

The E8 residual branch and its four matched controls are now implemented and unit-tested (Phase B3), including two tests running the actual real E8 kernel locally. **No E8-vs-controls statistical comparison on any benchmark dataset was run in this campaign.** Any future claim that E8's specific structure causes a measured gain requires running all five variants under matched seeds/schedule/compute and applying the preregistered statistical test (`preregistration.md` §8) — not attempted here, and explicitly Open in the claim matrix (CLM_KGR_018).

6 Scope of what this campaign's Phase C evidence actually is

`campaigns/gate12/tier0/`: A0 (TuckER-only) at full dataset scale, 2 seeds, both FB15K-237 and WN18RR; A3 (TuckER + path + seionic) on a 5,000-triple subsample, 2 seeds, both datasets. This is **screening-tier, not confirmatory** — 2 seeds is explicitly insufficient for any variance claim (reported per-seed, never as mean±std as if that were meaningful), the ablation ladder covers only 2 of the preregistered 13 points, and no dataset was run at the full preregistered epoch budget for A3. See `04_gate12_confirmatory_benchmark_report.tex` for the numbers themselves, reported with this scope caveat repeated, not just stated once here.

7 What must not be claimed from this campaign

- No SOTA claim. No protocol-compatible external-literature comparison was performed.
- No claim that the path reasoner, seionic branch, projection, rank controller, or E8 residual *causally improves* MRR on any benchmark dataset — Tier 0's A0-vs-A3 comparison is 2-seed, reduced-scale, and (per §2) confounded by the gate possibly never opening.
- No certified compression claim for any model using the path reasoner.
- No claim about GPU being faster than CPU for this architecture in general.